

```
>>ionic g page pages/input --spec=false
```

En archivo input.module.ts

```
import { NgModule } from '@angular/core';
import { CommonModule } from '@angular/common';
import { FormsModule } from '@angular/forms';

import { IonicModule } from '@ionic/angular';

import { InputPageRoutingModule } from './input-routing.module';

import { InputPage } from './input.page';
import { ComponentsModule } from '../components/components.module';

@NgModule({
  imports: [
    CommonModule,
    FormsModule,
    IonicModule,
    InputPageRoutingModule,
    ComponentsModule
  ],
  declarations: [InputPage]
})
export class InputPageModule { }
```

En archivo input.page.html

```
<app-header [titulo]="nombre || 'Input Forms'"></app-header>

<ion-content class="ion-padding">

  <ion-list>

    <ion-list-header>
      <ion-label>Input normal</ion-label>
    </ion-list-header>
```

```

    <ion-item>
      <ion-label>Nombre:</ion-label>
      <ion-input type="text" placeholder="Awesome Input"
[(ngModel)]= "nombre"></ion-input>
    </ion-item>

</ion-list>

<form #formulario="ngForm" (ngSubmit)="onSubmit(formulario)">

  <ion-list-header>
    <ion-label>Formulario válido: {{ formulario.valid }}</ion-label>
  </ion-list-header>

  <ion-item>
    <ion-label>Email:</ion-label>
    <ion-input type="email" placeholder="Email" name="email"
[(ngModel)]= "usuario.email"
      pattern="^([a-zA-Z0-9_\-\.]+)@([a-zA-Z0-9_\-\.]+)\.([a-zA-
Z]{2,5})$" required></ion-input>
  </ion-item>

  <ion-item>
    <ion-label>Password:</ion-label>
    <ion-input type="password" name="password"
[(ngModel)]= "usuario.password" required placeholder="Contraseña">
  </ion-input>
  </ion-item>

  <ion-button [disabled]="formulario.invalid" type="submit"
expand="block">
    Login
  </ion-button>

</form>

</ion-content>

```

En archivo input.page.ts

```

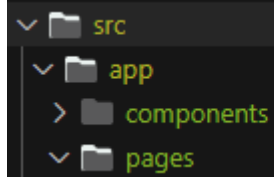
import { Component, OnInit } from '@angular/core';
import { NgForm } from '@angular/forms';

@Component({
  selector: 'app-input',
  templateUrl: './input.page.html',
  styleUrls: ['./input.page.scss'],
})
export class InputPage implements OnInit {
  nombre: string = 'Fernando';
  usuario = {
    email: '',
    password: ''
  }
  constructor() { }

  ngOnInit() {
  }

  onSubmit(formulario: NgForm) {
    console.log('submit');
    console.log(this.usuario);
    console.log(formulario);
  }
}

```



crear servicio para empezar a centralizar procesos.
 >>ionic g service services/data

app.module.ts actual

```

@NgModule({
  declarations: [AppComponent],
  imports: [
    BrowserModule,
    IonicModule.forRoot(),
    AppRoutingModule
  ],
  providers: [{ provide: RouteReuseStrategy, useClass: IonicRouteStrategy }],
  bootstrap: [AppComponent],
})
export class AppModule {}

```

Cambios:

```

import { NgModule } from '@angular/core';
import { BrowserModule } from '@angular/platform-browser';
import { RouteReuseStrategy } from '@angular/router';
import { HttpClientModule } from '@angular/common/http';

import { IonicModule, IonicRouteStrategy } from '@ionic/angular';

import { AppComponent } from './app.component';
import { AppRoutingModule } from './app-routing.module';

@NgModule({
  declarations: [AppComponent],
  imports: [
    BrowserModule,
    IonicModule.forRoot(),
    AppRoutingModule,
    HttpClientModule
  ],
  providers: [{ provide: RouteReuseStrategy, useClass: IonicRouteStrategy }],
  bootstrap: [AppComponent],
})
export class AppModule { }

```

▼ Menu

ion-menu

ion-menu-button

ion-menu-toggle

ion-split-pane

app.component.html

```
<ion-app>

  <ion-menu side="start" menuId="first" contentId="main">
    <ion-header>
      <ion-toolbar color="primary">
        <ion-title>Start Menu</ion-title>
      </ion-toolbar>
    </ion-header>
    <ion-content>
      <ion-list>
        <ion-item>Menu Item</ion-item>
        <ion-item>Menu Item</ion-item>
        <ion-item>Menu Item</ion-item>
        <ion-item>Menu Item</ion-item>
        <ion-item>Menu Item</ion-item>
      </ion-list>
    </ion-content>
  </ion-menu>

  <ion-router-outlet id="main"></ion-router-outlet>
</ion-app>
```

inicio.page.html

```
<ion-header>
  <ion-toolbar>

    <ion-buttons slot="start">
      <ion-button (click)="mostrarMenu()" color="primary">
        <ion-icon slot="icon-only" name="menu-outline"></ion-icon>
        Click me
      </ion-button>
    </ion-buttons>

    <ion-title color="primary">Componentes</ion-title>
  </ion-toolbar>
</ion-header>

<ion-content>

  <ion-list>
```

```

    <ion-item *ngFor="let c of componentes" [routerLink]="c.redirectTo"
detail>
    <ion-icon [name]="c.icon" slot="start" color="primary"></ion-icon>
    {{c.name}}
  </ion-item>
</ion-list>

```

inicio.page.html

otra opción de ion-button

```

<ion-header>
  <ion-toolbar>

    <ion-buttons slot="start">
      <ion-menu-button menu="first" color="primary"></ion-menu-button>
    </ion-buttons>

    <ion-title color="primary">Componentes</ion-title>
  </ion-toolbar>
</ion-header>

<ion-content>

  <ion-list>
    <ion-item *ngFor="let c of componentes" [routerLink]="c.redirectTo"
detail>
      <ion-icon [name]="c.icon" slot="start" color="primary"></ion-icon>
      {{c.name}}
    </ion-item>
  </ion-list>
</ion-content>

```

inicio.page.ts

```

import { Component, OnInit } from '@angular/core';
import { MenuController } from '@ionic/angular';
import { ComponentsModule } from '../components/components.module';

//crear una interface para definir propiedades que el componente lista va
a tener
interface Componente {
  icon: string;
  name: string;
  redirectTo: string;
}

```

```

@Component({
  selector: 'app-inicio',
  templateUrl: './inicio.page.html',
  styleUrls: ['./inicio.page.scss'],
})
export class InicioPage implements OnInit {

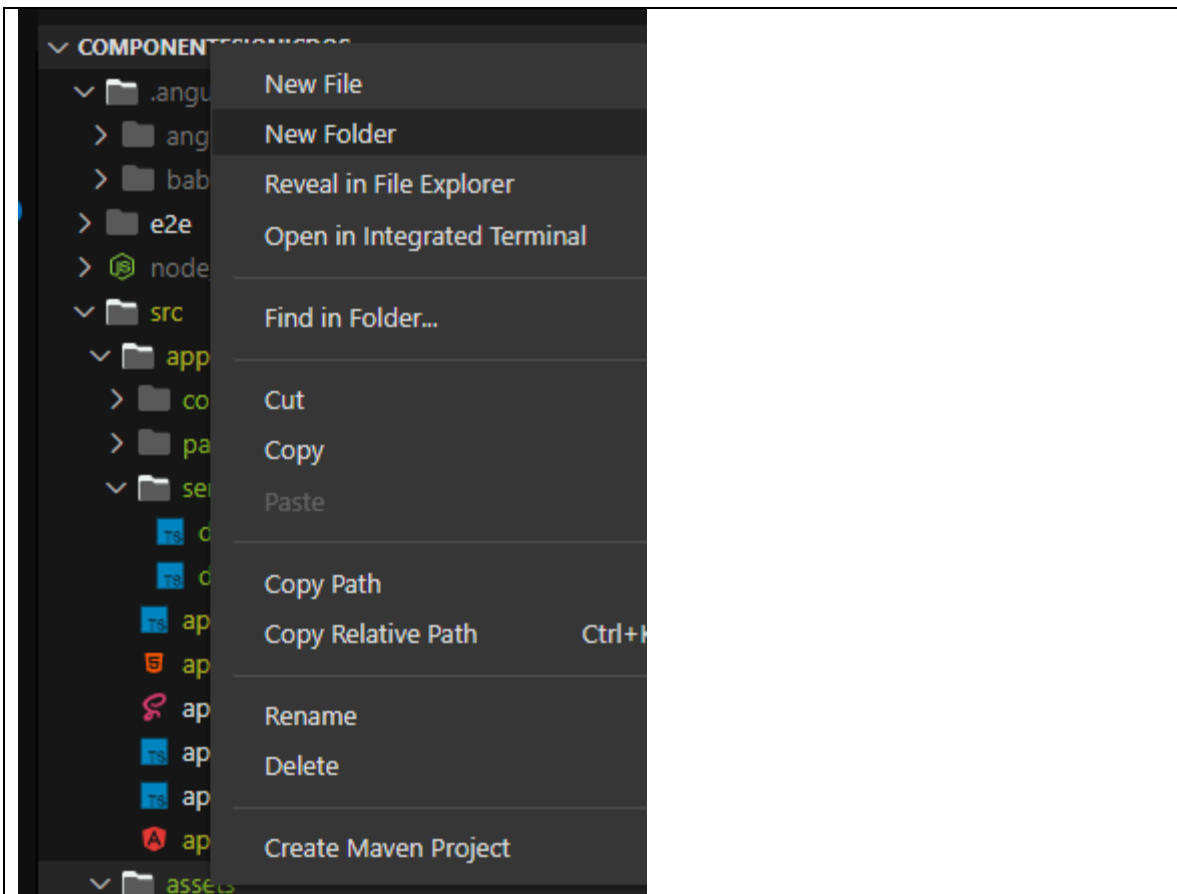
  //crear nuevo listado

  //componentes: any[] = [];
  componentes: Componente[] = [
    {
      icon: 'american-football-outline',
      name: 'Action Sheet',
      redirectTo: '/action-sheet'
    },
    {
      icon: 'alert-circle-outline',
      name: 'Alert',
      redirectTo: '/alert'
    },
    {
      icon: 'beaker-outline',
      name: 'Avatar',
      redirectTo: '/avatar'
    }
  ];

  constructor(private menuCtrl: MenuController) { }

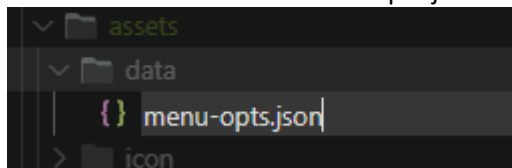
  ngOnInit() {
  }
  mostrarMenu() {
    this.menuCtrl.open('first');
  }
}

```



Crear carpeta Data

En data crear el archivo menu-opts.json



La información se copia del archivo inicio.page.ts

Menu-opts.json

```
[
  {
    "icon": "american-football-outline",
    "name": "Action Sheet",
    "redirectTo": "/action-sheet"
  },
  {
    "icon": "alert-circle-outline",
    "name": "Alert",
    "redirectTo": "/alert"
  },
  {
    "icon": "beaker-outline",
```

```
    "name": "Avatar",
    "redirectTo": "/avatar"
  }
]
```

En el archivo inicio.page.ts
quedaría el arreglo:

```
componentes: Componente[] = [];
```

```
import { Component, OnInit } from '@angular/core';
import { MenuController } from '@ionic/angular';
import { Observable } from 'rxjs';
import { ComponentsModule } from '../components/components.module';
import { DataService } from '../services/data.service';

//crear una interface para definir propiedades que el componente lista va
a tener
interface Componente {
  icon: string;
  name: string;
  redirectTo: string;
}

@Component({
  selector: 'app-inicio',
  templateUrl: './inicio.page.html',
  styleUrls: ['./inicio.page.scss'],
})
export class InicioPage implements OnInit {

  //crear nuevo listado

  //componentes: any[] = [];
  componentes: Componente[] = [];
  componentes: Observable<Componente[]>;

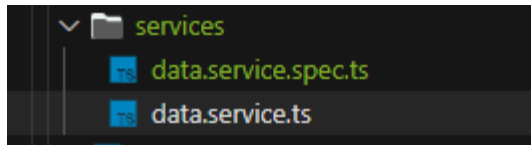
  constructor(private menuCtrl: MenuController,
  ) { }

  ngOnInit() {
  }

  mostrarMenu() {
```

```
    this.menuCtrl.open('first');  
  }  
}
```

Ira:



En el archivo data.service.ts

```
import { Injectable, Component } from '@angular/core';  
import { HttpClient } from '@angular/common/http';  
import { Componente } from '../interfaces/interfaces';  
  
@Injectable({  
  providedIn: 'root'  
})  
export class DataService {  
  
  constructor(private http: HttpClient) { }  
  
  getMenuOpts() {  
    return this.http.get<Componente[]>('/assets/data/menu-opts.json');  
  }  
}
```

En app crear la carpeta interfaces y crear interfaces/ interfaces.ts

interfaces.ts

```
export interface Componente {  
  icon: string;  
  name: string;  
  redirectTo: string;  
}
```

inicio.page.html

```

<ion-header>
  <ion-toolbar>

    <ion-buttons slot="start">
      <ion-menu-button menu="first" color="primary"></ion-menu-button>
    </ion-buttons>

    <ion-title color="primary">Componentes</ion-title>
  </ion-toolbar>
</ion-header>

<ion-content>

  <ion-list>
    <ion-item *ngFor="let c of componentes | async"
[routerLink]="c.redirectTo" detail>
      <ion-icon [name]="c.icon" slot="start" color="primary"></ion-icon>
      {{c.name}}
    </ion-item>
  </ion-list>

</ion-content>

```

Hacer cambios en inicio.page.ts

```

import { Component, OnInit } from '@angular/core';
import { MenuController } from '@ionic/angular';
import { Observable } from 'rxjs';
import { ComponentsModule } from '../components/components.module';
import { DataService } from '../services/data.service';

//crear una interface para definir propiedades que el componente lista va
a tener
interface Componente {
  icon: string;
  name: string;
  redirectTo: string;
}

@Component({

```

```

selector: 'app-inicio',
templateUrl: './inicio.page.html',
styleUrls: ['./inicio.page.scss'],
}))
export class InicioPage implements OnInit {

  componentes: Observable<Componente[]>;

  constructor(private menuCtrl: MenuController,
    private dataService: DataService) { }

  ngOnInit() {
    this.componentes = this.dataService.getMenuOpts();
  }
  mostrarMenu() {
    this.menuCtrl.open('first');
  }
}

```

inicio.page.html

```

<ion-header>
  <ion-toolbar>

    <ion-buttons slot="start">
      <ion-menu-button menu="first" color="primary"></ion-menu-button>
    </ion-buttons>

    <ion-title color="primary">Componentes</ion-title>
  </ion-toolbar>
</ion-header>

<ion-content>

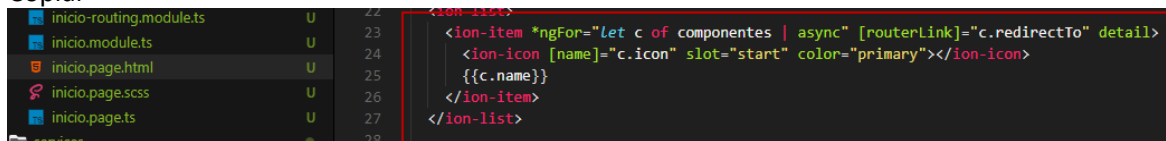
  <ion-list>
    <ion-item *ngFor="let c of componentes | async"
[routerLink]="c.redirectTo" detail>
      <ion-icon [name]="c.icon" slot="start" color="primary"></ion-icon>
      {{c.name}}
    </ion-item>
  </ion-list>
</ion-content>

```

```
</ion-list>

</ion-content>
```

Copiar



```
<ion-list>
  <ion-item *ngFor="let c of componentes | async" [routerLink]="c.redirectTo" detail>
    <ion-icon [name]="c.icon" slot="start" color="primary"></ion-icon>
    {{c.name}}
  </ion-item>
</ion-list>
```

app.component.html

```
<ion-list>
  <ion-item *ngFor="let c of componentes | async"
[routerLink]="c.redirectTo" detail>
    <ion-icon [name]="c.icon" slot="start" color="primary"></ion-icon>
    {{c.name}}
  </ion-item>
</ion-list>
```

Cambios: app.component.html

```
<ion-app>

  <ion-menu side="start" menuId="first" contentId="main">
    <ion-header>
      <ion-toolbar color="primary">
        <ion-title>Menú</ion-title>
      </ion-toolbar>
    </ion-header>
    <ion-content>
      <ion-list>

        <ion-menu-toggle *ngFor="let c of componentes | async">
          <ion-item [routerLink]="c.redirectTo" detail>
            <ion-icon [name]="c.icon" slot="start" color="primary"></ion-
icon>
            {{c.name}}
          </ion-item>
        </ion-menu-toggle>
```

```

        </ion-list>
      </ion-content>
    </ion-menu>

    <ion-router-outlet id="main"></ion-router-outlet>
  </ion-app>

```

Ir a inicio.page.ts

Y copiar:

```

componentes: Observable<Componente[]>;
constructor(private menuCtrl: MenuController,
  private dataService: DataService) { }

ngOnInit() {
  this.componentes = this.dataService.getMenuOpts();
}

```

app.component.ts, quedaría así:

```

import { Component } from '@angular/core';
import { Observable } from 'rxjs/internal/Observable';
import { Componente } from '../interfaces/interfaces';
import { DataService } from '../services/data.service';

@Component({
  selector: 'app-root',
  templateUrl: 'app.component.html',
  styleUrls: ['app.component.scss'],
})
export class AppComponent {
  componentes: Observable<Componente[]>;

  ngOnInit() {
    this.componentes = this.dataService.getMenuOpts();
  }
}

```

```

constructor(private dataService: DataService) {

}

initializeApp() {
  this.componentes = this.dataService.getMenuOpts();
}
}

```

En app.component.html quedaría así:

```

<ion-app>

  <ion-menu side="start" menuId="first" contentId="main">
    <ion-header>
      <ion-toolbar color="primary">
        <ion-title>Menú</ion-title>
      </ion-toolbar>
    </ion-header>
    <ion-content>
      <ion-list>

        <ion-menu-toggle *ngFor="let c of componentes | async">
          <ion-item [routerLink]="c.redirectTo" detail>
            <ion-icon [name]="c.icon" slot="start" color="primary"></ion-
icon>
            {{c.name}}
          </ion-item>
        </ion-menu-toggle>

      </ion-list>
    </ion-content>
  </ion-menu>

  <ion-router-outlet id="main"></ion-router-outlet>
</ion-app>

```

